

A Self-Contained Convergence Proof of a Short-Step IPM for Box-Constrained Linear Programs

Pedagogical English rewrite

July 1, 2026

Abstract

This note gives a self-contained convergence proof for a short-step primal-dual interior-point method for the linear program

$$\min_x c^\top x \quad \text{subject to} \quad Ax \geq b, \quad 0 \leq x \leq 1.$$

The proof is written for a reader who may not already know optimization. The main idea is simple: the algorithm keeps every inequality strictly away from the boundary, follows the “central path”, and decreases a number called the duality measure. The technical heart of the proof is the normalized change of variables

$$u_i = \Delta z_i \sqrt{\lambda_i / z_i}, \quad v_i = \Delta \lambda_i \sqrt{z_i / \lambda_i},$$

where each pair (z_i, λ_i) is a primal slack and its dual multiplier. This scaling is not a trick: it is the unique natural way to control both the first-order complementarity error and the second-order product term after a Newton update. The note also explains how this theoretical short-step IPM differs from modern practical IPM implementations and from the IPM-MPNN perspective of Qian, Chetelat, and Morris.

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1 Reader's map: what we will prove

We study the problem

$$\min_x c^\top x \quad \text{subject to} \quad Ax \geq b, \quad 0 \leq x \leq 1. \quad (1)$$

The vector $x \in \mathbb{R}^n$ is the decision variable. The matrix $A \in \mathbb{R}^{m \times n}$ and vectors b, c are data. The phrase “linear program” means that both the objective $c^\top x$ and all constraints are linear.

The interior-point method (IPM) does not walk on the boundary of the feasible region. Instead, it keeps all inequalities strict:

$$Ax > b, \quad x > 0, \quad x < 1.$$

Then it gradually moves toward optimality while staying inside the feasible region.

The proof has four goals.

1. Define the primal variables, dual variables, and the duality gap.
2. Define a Newton direction that points toward the central path.
3. Prove that a small universal step keeps the iterate in a safe central neighborhood.
4. Prove that the duality gap decreases geometrically, hence the objective value converges to the optimum.

The whole proof is algebraic. Once the right variables are introduced, the convergence argument reduces to controlling the products

$$\text{slack} \times \text{dual multiplier}.$$

2 The primal problem and slack variables

The original constraints are

$$Ax \geq b, \quad x \geq 0, \quad x \leq 1.$$

It is useful to convert each inequality into a nonnegative “slack” variable.

Definition 2.1 (Slack variables). Define

$$r := Ax - b \in \mathbb{R}^m, \quad q := \mathbf{1} - x \in \mathbb{R}^n.$$

Then the constraints are equivalent to

$$r \geq 0, \quad x \geq 0, \quad q \geq 0, \quad r = Ax - b, \quad q = \mathbf{1} - x.$$

Strict interior feasibility means

$$r > 0, \quad x > 0, \quad q > 0.$$

The box constraint $0 \leq x \leq 1$ is therefore represented by two nonnegative objects: x itself, measuring distance from the lower bound 0, and $q = 1 - x$, measuring distance from the upper bound 1.

3 The dual problem and the duality gap

The dual problem gives lower bounds on the primal optimum. We derive it slowly because the signs matter.

The primal problem is

$$\min_x c^\top x \quad \text{subject to} \quad Ax - b \geq 0, \quad x \geq 0, \quad \mathbf{1} - x \geq 0. \quad (\text{P})$$

For constraints written as “ ≥ 0 ” in a minimization problem, we subtract nonnegative multipliers in the Lagrangian. Let

$$w \geq 0 \quad \text{for } Ax - b \geq 0, \quad s \geq 0 \quad \text{for } x \geq 0, \quad k \geq 0 \quad \text{for } \mathbf{1} - x \geq 0.$$

The Lagrangian is

$$L(x; w, s, k) = c^\top x - w^\top (Ax - b) - s^\top x - k^\top (\mathbf{1} - x). \quad (2)$$

Rearranging the terms involving x gives

$$L(x; w, s, k) = (c - A^\top w - s + k)^\top x + b^\top w - \mathbf{1}^\top k. \quad (3)$$

If

$$c - A^\top w - s + k = 0, \quad w, s, k \geq 0, \quad (4)$$

then the x -dependent term disappears, and the Lagrangian value is always

$$b^\top w - \mathbf{1}^\top k.$$

For any feasible x , since the subtracted constraint terms in (2) are nonpositive, we have

$$L(x; w, s, k) \leq c^\top x.$$

Therefore $b^\top w - \mathbf{1}^\top k$ is a lower bound on every feasible primal objective value. The dual problem is

$$\max_{w, s, k} b^\top w - \mathbf{1}^\top k \quad \text{subject to} \quad c - A^\top w - s + k = 0, \quad w, s, k \geq 0. \quad (\text{D})$$

Lemma 3.1 (Duality gap identity). *Suppose x is primal feasible and (w, s, k) is dual feasible. Let*

$$r = Ax - b, \quad q = \mathbf{1} - x.$$

Then

$$c^\top x - (b^\top w - \mathbf{1}^\top k) = x^\top s + q^\top k + r^\top w. \quad (5)$$

Proof. Dual feasibility means

$$c = A^\top w + s - k.$$

Substitute this into the primal objective:

$$\begin{aligned} c^\top x - (b^\top w - \mathbf{1}^\top k) &= (A^\top w + s - k)^\top x - b^\top w + \mathbf{1}^\top k \\ &= w^\top (Ax - b) + s^\top x + k^\top (\mathbf{1} - x) \\ &= r^\top w + x^\top s + q^\top k. \end{aligned}$$

This proves the identity. $\square$

The right-hand side of (5) is a sum of nonnegative products. Hence, if we can make all products

$$x_i s_i, \quad q_i k_i, \quad r_j w_j$$

small, then the primal objective value is close to the best possible value.

4 Barrier viewpoint and the central path

An interior-point method avoids the boundary by adding logarithmic penalties. The barrier problem is

$$\min_{0 < x < 1, Ax > b} \phi_\mu(x) := c^\top x - \mu \sum_{j=1}^m \log((Ax - b)_j) - \mu \sum_{i=1}^n \log x_i - \mu \sum_{i=1}^n \log(1 - x_i), \quad (6)$$

where $\mu > 0$ is a parameter.

The logarithms have a simple role. When x_i approaches 0, $\log x_i \rightarrow -\infty$, so $-\log x_i \rightarrow +\infty$. The objective therefore strongly discourages approaching the boundary. The same happens near $x_i = 1$ and near $Ax = b$.

The first-order condition for (6) is

$$c - \mu A^\top r^{-1} - \mu x^{-1} + \mu q^{-1} = 0, \quad (7)$$

where inverse is coordinatewise. Define

$$w_j = \frac{\mu}{r_j}, \quad s_i = \frac{\mu}{x_i}, \quad k_i = \frac{\mu}{q_i}. \quad (8)$$

Then (7) becomes

$$c - A^\top w - s + k = 0.$$

Moreover,

$$r_j w_j = \mu, \quad x_i s_i = \mu, \quad q_i k_i = \mu.$$

Definition 4.1 (Central path). The central path is the set of strictly feasible primal-dual points satisfying

$$\begin{aligned}
r &= Ax - b, & r &> 0, \\
q &= \mathbf{1} - x, & x &> 0, \quad q > 0, \\
c - A^\top w - s + k &= 0, & w, s, k &> 0, \\
Xs &= \mu \mathbf{1}, \\
Qk &= \mu \mathbf{1}, \\
Rw &= \mu \mathbf{1},
\end{aligned} \tag{9}$$

where

$$X = \text{diag}(x), \quad Q = \text{diag}(q), \quad R = \text{diag}(r), \quad S = \text{diag}(s), \quad K = \text{diag}(k), \quad W = \text{diag}(w).$$

On the central path, every complementarity product equals the same number μ . As $\mu \downarrow 0$, all products approach zero, so the duality gap approaches zero.

5 The duality measure and the central neighborhood

In practice, iterates do not lie exactly on the central path. Instead, we measure their average complementarity.

Definition 5.1 (Duality measure). Let

$$N := 2n + m. \tag{10}$$

There are n products $x_i s_i$, n products $q_i k_i$, and m products $r_j w_j$, hence N products in total. Define

$$\mu := \frac{x^\top s + q^\top k + r^\top w}{N}. \tag{11}$$

By Lemma 3.1, the duality gap is exactly $N\mu$.

Definition 5.2 (Central neighborhood). Fix a constant $\gamma \in (0, 1)$. The central neighborhood $\mathcal{C}_\gamma$ consists of strictly primal-dual feasible points satisfying

$$\min \left\{ \min_i x_i s_i, \min_i q_i k_i, \min_j r_j w_j \right\} \geq \gamma \mu. \tag{12}$$

This definition says that no product is allowed to be much smaller than the average product μ . The neighborhood is a safety region: it prevents the algorithm from becoming too close to one boundary while other coordinates remain far away.

Assumption 5.3 (Clean feasible-start assumption). *The proof assumes that we are given an initial point in $\mathcal{C}_\gamma$. In other words, the initial point is strictly primal feasible, strictly dual feasible, and centrally balanced in the sense of (12).*

This is a standard simplifying assumption for short-step theory. Practical solvers usually do not require such an initial point; see Section 14.

6 The Newton direction

At a current point in $\mathcal{C}_\gamma$, let μ be defined by (11). Choose a constant $\sigma \in (0, 1)$. The role of σ is to choose the next complementarity target $\sigma\mu$, which is smaller than the current average μ .

The Newton direction

$$(\Delta x, \Delta q, \Delta r, \Delta w, \Delta s, \Delta k)$$

is defined by the linear system

$$\begin{aligned} A\Delta x - \Delta r &= 0, \\ \Delta x + \Delta q &= 0, \\ -A^\top \Delta w - \Delta s + \Delta k &= 0, \\ S\Delta x + X\Delta s &= \sigma\mu\mathbf{1} - Xs, \\ K\Delta q + Q\Delta k &= \sigma\mu\mathbf{1} - Qk, \\ W\Delta r + R\Delta w &= \sigma\mu\mathbf{1} - Rw. \end{aligned} \tag{13}$$

The first three equations ensure that the equality constraints stay true after a step. The last three equations are the Newton linearizations of

$$Xs = \sigma\mu\mathbf{1}, \quad Qk = \sigma\mu\mathbf{1}, \quad Rw = \sigma\mu\mathbf{1}.$$

After computing the direction, we update by

$$(x, q, r, w, s, k)^+ = (x, q, r, w, s, k) + \alpha(\Delta x, \Delta q, \Delta r, \Delta w, \Delta s, \Delta k), \tag{14}$$

where $\alpha \in (0, 1]$ is the step size.

7 Why the variables u_i and v_i are natural

This section is the conceptual core of the proof.

Every complementarity pair has the same form:

$$z_i > 0, \quad \lambda_i > 0, \quad p_i := z_i\lambda_i.$$

In our problem the pairs are

$$(z, \lambda) = ((x, q, r), (s, k, w)).$$

For each pair, the Newton complementarity equation says

$$\lambda_i \Delta z_i + z_i \Delta \lambda_i = \sigma\mu - p_i. \tag{15}$$

After a step of length α , the product becomes

$$\begin{aligned} (z_i + \alpha \Delta z_i)(\lambda_i + \alpha \Delta \lambda_i) &= z_i \lambda_i + \alpha(\lambda_i \Delta z_i + z_i \Delta \lambda_i) + \alpha^2 \Delta z_i \Delta \lambda_i \\ &= (1 - \alpha)p_i + \alpha\sigma\mu + \alpha^2 \Delta z_i \Delta \lambda_i. \end{aligned} \tag{16}$$

The first two terms are controlled by the Newton equation. The difficult term is the second-order term

$$\Delta z_i \Delta \lambda_i.$$

We therefore introduce

$$u_i := \Delta z_i \sqrt{\frac{\lambda_i}{z_i}}, \quad v_i := \Delta \lambda_i \sqrt{\frac{z_i}{\lambda_i}}. \tag{17}$$

These variables satisfy two identities:

$$u_i + v_i = \frac{\lambda_i \Delta z_i + z_i \Delta \lambda_i}{\sqrt{z_i \lambda_i}} = \frac{\sigma \mu - p_i}{\sqrt{p_i}}, \quad u_i v_i = \Delta z_i \Delta \lambda_i. \quad (18)$$

Thus $u_i + v_i$ is the normalized first-order error, and $u_i v_i$ is exactly the second-order product error. This explains the particular scaling

$$u_i = \Delta x_i \sqrt{\frac{s_i}{x_i}}$$

for the pair (x_i, s_i) : it balances the primal quantity x_i and the dual quantity s_i so that both the first-order and second-order expressions become simple.

For the box-constrained LP, the complete definitions are

$$\begin{aligned} u_{x,i} &= \Delta x_i \sqrt{\frac{s_i}{x_i}}, & v_{x,i} &= \Delta s_i \sqrt{\frac{x_i}{s_i}}, \\ u_{q,i} &= \Delta q_i \sqrt{\frac{k_i}{q_i}}, & v_{q,i} &= \Delta k_i \sqrt{\frac{q_i}{k_i}}, \\ u_{r,j} &= \Delta r_j \sqrt{\frac{w_j}{r_j}}, & v_{r,j} &= \Delta w_j \sqrt{\frac{r_j}{w_j}}. \end{aligned}$$

Stack all u variables into one vector $U \in \mathbb{R}^N$, and all v variables into one vector $V \in \mathbb{R}^N$.

8 Two algebraic lemmas

Lemma 8.1 (Orthogonality). *The Newton direction satisfies*

$$U^\top V = 0. \quad (19)$$

Consequently,

$$\|U + V\|_2^2 = \|U\|_2^2 + \|V\|_2^2. \quad (20)$$

Proof. By the definitions of U and V ,

$$U^\top V = \Delta x^\top \Delta s + \Delta q^\top \Delta k + \Delta r^\top \Delta w.$$

Since $q = 1 - x$, the Newton direction satisfies $\Delta q = -\Delta x$. Hence

$$U^\top V = \Delta x^\top \Delta s - \Delta x^\top \Delta k + \Delta r^\top \Delta w.$$

From the stationarity equation in (13),

$$-A^\top \Delta w - \Delta s + \Delta k = 0,$$

so

$$\Delta s - \Delta k = -A^\top \Delta w.$$

Multiplying by $\Delta x^\top$ gives

$$\Delta x^\top \Delta s - \Delta x^\top \Delta k = -\Delta x^\top A^\top \Delta w.$$

The primal equality direction gives $A \Delta x = \Delta r$. Therefore

$$-\Delta x^\top A^\top \Delta w = -(A \Delta x)^\top \Delta w = -\Delta r^\top \Delta w.$$

Substituting this into the expression for $U^\top V$ yields $U^\top V = 0$. The norm identity follows from expanding

$$\|U + V\|_2^2 = \|U\|_2^2 + 2U^\top V + \|V\|_2^2.$$

□

Lemma 8.2 (Bound on the normalized residual). *Assume the current point is in $\mathcal{C}_\gamma$. Let*

$$C_{\gamma,\sigma} := 1 - 2\sigma + \frac{\sigma^2}{\gamma}.$$

Then

$$\|U + V\|_2^2 \leq C_{\gamma,\sigma} N\mu. \quad (21)$$

Proof. Let $p_i = z_i \lambda_i$ be the i -th complementarity product. By (18),

$$u_i + v_i = \frac{\sigma\mu - p_i}{\sqrt{p_i}}.$$

Therefore

$$\begin{aligned} \|U + V\|_2^2 &= \sum_{i=1}^N \frac{(\sigma\mu - p_i)^2}{p_i} \\ &= \sum_{i=1}^N \left(p_i - 2\sigma\mu + \frac{\sigma^2\mu^2}{p_i} \right). \end{aligned}$$

Because $\sum_i p_i = N\mu$ and because the central neighborhood gives $p_i \geq \gamma\mu$ for every i ,

$$\sum_{i=1}^N \frac{1}{p_i} \leq \frac{N}{\gamma\mu}.$$

Hence

$$\|U + V\|_2^2 \leq N\mu - 2\sigma N\mu + \frac{\sigma^2\mu^2 N}{\gamma\mu} = \left(1 - 2\sigma + \frac{\sigma^2}{\gamma} \right) N\mu.$$

□

9 A safe step keeps the iterate in the central neighborhood

This is the main technical step. We show that if the current iterate is safe, then taking a sufficiently small step remains safe. The step size may be conservative, but it is uniform: it depends only on N , γ , and σ , not on the numerical values of the LP data.

Proposition 9.1 (Safe short step). *Let the current point be in $\mathcal{C}_\gamma$. Define*

$$C := C_{\gamma,\sigma} = 1 - 2\sigma + \frac{\sigma^2}{\gamma}.$$

For every

$$0 \leq \alpha \leq \bar{\alpha} := \min \left\{ 1, \frac{2\sigma(1-\gamma)}{CN}, \frac{1}{2} \sqrt{\frac{\gamma}{CN}} \right\}, \quad (22)$$

the updated point (14) remains in $\mathcal{C}_\gamma$.

Proof. We prove three facts: equality feasibility, positivity, and centrality.

Step 1: equality feasibility. If $r = Ax - b$, then

$$r + \alpha \Delta r = Ax - b + \alpha A \Delta x = A(x + \alpha \Delta x) - b,$$

because $A \Delta x - \Delta r = 0$. Hence the equality $r = Ax - b$ is preserved.

If $q = 1 - x$, then

$$q + \alpha \Delta q = 1 - x - \alpha \Delta x = 1 - (x + \alpha \Delta x),$$

because $\Delta q = -\Delta x$. Hence $q = 1 - x$ is preserved.

If $c - A^\top w - s + k = 0$, then

$$\begin{aligned} c - A^\top(w + \alpha \Delta w) - (s + \alpha \Delta s) + (k + \alpha \Delta k) &= c - A^\top w - s + k + \alpha(-A^\top \Delta w - \Delta s + \Delta k) \\ &= 0. \end{aligned}$$

Thus dual equality feasibility is also preserved.

Step 2: positivity. For a generic pair (z_i, λ_i) , by the definition of u_i ,

$$\left| \frac{\Delta z_i}{z_i} \right| = \frac{|u_i|}{\sqrt{z_i \lambda_i}} = \frac{|u_i|}{\sqrt{p_i}}.$$

The central neighborhood gives $p_i \geq \gamma \mu$. Lemmas 8.1 and 8.2 imply

$$|u_i| \leq \|U\|_2 \leq \|U + V\|_2 \leq \sqrt{CN\mu}.$$

Therefore

$$\left| \frac{\Delta z_i}{z_i} \right| \leq \sqrt{\frac{CN}{\gamma}}.$$

The same argument gives

$$\left| \frac{\Delta \lambda_i}{\lambda_i} \right| \leq \sqrt{\frac{CN}{\gamma}}.$$

If

$$\alpha \leq \frac{1}{2} \sqrt{\frac{\gamma}{CN}},$$

then

$$z_i + \alpha \Delta z_i \geq \frac{1}{2} z_i > 0, \quad \lambda_i + \alpha \Delta \lambda_i \geq \frac{1}{2} \lambda_i > 0.$$

Thus all variables remain strictly positive.

Step 3: centrality. Let

$$p_i(\alpha) := (z_i + \alpha \Delta z_i)(\lambda_i + \alpha \Delta \lambda_i).$$

By (16),

$$p_i(\alpha) = (1 - \alpha)p_i + \alpha \sigma \mu + \alpha^2 u_i v_i. \tag{23}$$

Summing over all i and using $\sum_i u_i v_i = U^\top V = 0$, we obtain

$$\sum_{i=1}^N p_i(\alpha) = (1 - \alpha)N\mu + \alpha \sigma N\mu, \tag{24}$$

so the new duality measure is

$$\mu(\alpha) = (1 - \alpha(1 - \sigma))\mu. \tag{25}$$

We need to prove that every new product satisfies

$$p_i(\alpha) \geq \gamma\mu(\alpha).$$

Using $p_i \geq \gamma\mu$ and $|u_i v_i| \leq (u_i^2 + v_i^2)/2$, we get

$$\begin{aligned} p_i(\alpha) &\geq (1 - \alpha)\gamma\mu + \alpha\sigma\mu - \frac{\alpha^2}{2}(u_i^2 + v_i^2) \\ &\geq (1 - \alpha)\gamma\mu + \alpha\sigma\mu - \frac{\alpha^2}{2}(\|U\|_2^2 + \|V\|_2^2) \\ &= (1 - \alpha)\gamma\mu + \alpha\sigma\mu - \frac{\alpha^2}{2}\|U + V\|_2^2 \\ &\geq (1 - \alpha)\gamma\mu + \alpha\sigma\mu - \frac{\alpha^2}{2}CN\mu. \end{aligned}$$

It is enough to require this lower bound to be at least

$$\gamma(1 - \alpha(1 - \sigma))\mu.$$

After cancelling common terms, the sufficient condition becomes

$$\alpha\sigma(1 - \gamma) - \frac{\alpha^2 CN}{2} \geq 0,$$

which holds whenever

$$\alpha \leq \frac{2\sigma(1 - \gamma)}{CN}.$$

This is exactly one of the restrictions in (22). Therefore the updated point remains in the central neighborhood. $\square$

Corollary 9.2 (A simple lower bound on the step size). *There exists a constant $\kappa > 0$, depending only on γ and σ , such that every iteration can take a step with*

$$\alpha \geq \frac{\kappa}{N}. \tag{26}$$

Proof. From (22), define

$$\kappa := \min \left\{ 1, \frac{2\sigma(1 - \gamma)}{C}, \frac{1}{2} \sqrt{\frac{\gamma}{C}} \right\}.$$

Since $N \geq 1$, we have $1 \geq 1/N$ and $1/\sqrt{N} \geq 1/N$. Hence $\bar{\alpha} \geq \kappa/N$. $\square$

10 Convergence of the duality measure

We now prove that the average complementarity μ decreases geometrically.

Theorem 10.1 (Linear convergence of the duality measure). *Under Assumption 5.3, the short-step IPM generates iterates in $\mathcal{C}_\gamma$ for all t , and*

$$\mu_{t+1} \leq \left(1 - \frac{(1 - \sigma)\kappa}{N} \right) \mu_t, \tag{27}$$

where $\kappa > 0$ depends only on γ and σ . Therefore

$$\mu_t \leq \exp \left(-\frac{(1 - \sigma)\kappa}{N} t \right) \mu_0. \tag{28}$$

Proof. By Proposition 9.1, a safe step keeps the iterate in $\mathcal{C}_\gamma$. Hence the statement holds inductively for all iterations.

From (25), after a step of size α_t ,

$$\mu_{t+1} = (1 - \alpha_t(1 - \sigma))\mu_t.$$

By Corollary 9.2, we can take $\alpha_t \geq \kappa/N$. Therefore

$$\mu_{t+1} \leq \left(1 - \frac{(1 - \sigma)\kappa}{N}\right) \mu_t.$$

Iterating this inequality and using $1 - a \leq e^{-a}$ gives (28). $\square$

11 Convergence to optimality

Theorem 11.1 (Objective gap convergence). *Let $p^\star$ be the optimal value of the primal LP. Then the primal iterates satisfy*

$$0 \leq c^\top x_t - p^\star \leq N\mu_t. \quad (29)$$

Consequently, it is sufficient to take

$$t \geq \frac{N}{(1 - \sigma)\kappa} \log \left(\frac{N\mu_0}{\varepsilon} \right) \quad (30)$$

iterations to guarantee $c^\top x_t - p^\star \leq \varepsilon$.

Proof. The dual objective value $b^\top w_t - \mathbf{1}^\top k_t$ is a lower bound on the primal optimum $p^\star$. Therefore

$$0 \leq c^\top x_t - p^\star \leq c^\top x_t - (b^\top w_t - \mathbf{1}^\top k_t).$$

By the duality gap identity in Lemma 3.1,

$$c^\top x_t - (b^\top w_t - \mathbf{1}^\top k_t) = x_t^\top s_t + q_t^\top k_t + r_t^\top w_t = N\mu_t.$$

This proves (29). Combining it with Theorem 10.1, it is enough to require

$$N\mu_0 \exp \left(-\frac{(1 - \sigma)\kappa}{N} t \right) \leq \varepsilon,$$

which rearranges to (30). $\square$

Thus the iteration complexity is

$$O((2n + m) \log(1/\varepsilon)),$$

up to constants depending on γ , σ , and the initial duality measure.

12 Optional: convergence to the set of optimal solutions

The previous theorem proves objective convergence. Because the feasible set is bounded by $0 \leq x \leq 1$, objective convergence also implies that every limit point is optimal.

Proposition 12.1 (Distance to the optimal face). *Let*

$$X^* := \{x : Ax \geq b, \ 0 \leq x \leq 1, \ c^\top x = p^*\}.$$

Then

$$\text{dist}(x_t, X^*) \rightarrow 0.$$

Moreover, by Hoffman's error bound for linear inequalities, there exists a constant $H > 0$, depending only on A and c , such that

$$\text{dist}(x_t, X^*) \leq HN\mu_t. \quad (31)$$

Proof. The feasible set is closed and bounded because $0 \leq x \leq 1$. Hence it is compact. Theorem 11.1 gives $c^\top x_t \rightarrow p^*$. Therefore every convergent subsequence of x_t has a limit point that is feasible and has objective value p^* , hence lies in X^* . This implies $\text{dist}(x_t, X^*) \rightarrow 0$.

For the quantitative statement, write the optimal set as the polyhedron

$$X^* = \{x : Ax \geq b, \ x \geq 0, \ \mathbf{1} - x \geq 0, \ c^\top x \leq p^*\}.$$

The iterates already satisfy the first three groups of inequalities exactly. Thus their only possible violation of this polyhedron is the inequality $c^\top x \leq p^*$. Hoffman's error bound gives a constant $H > 0$ such that

$$\text{dist}(x_t, X^*) \leq H \max\{0, c^\top x_t - p^*\} \leq HN\mu_t,$$

where the last step uses (29). □

13 What changes because of the box constraint $0 \leq x \leq 1$?

Compared with the simpler LP

$$\min c^\top x \quad \text{subject to} \quad Ax \geq b, \quad x \geq 0,$$

the upper bound $x \leq 1$ creates four concrete changes.

1. **A new slack variable.** We must introduce $q = 1 - x$. The algorithm tracks both x and q .
2. **A new dual multiplier.** The upper bound has multiplier $k \geq 0$. The dual stationarity condition becomes

$$c - A^\top w - s + k = 0.$$

The plus sign in front of k is essential.

3. **A larger complementarity system.** The complementarity pairs are now

$$(x, s), \quad (q, k), \quad (r, w),$$

so the number of products is $N = 2n + m$ instead of $n + m$.

4. **Two-sided barrier curvature.** If one eliminates the other Newton variables, the x -block contains the diagonal matrix

$$X^{-1}S + Q^{-1}K.$$

The term $X^{-1}S$ comes from the lower wall $x > 0$, and the term $Q^{-1}K$ comes from the upper wall $x < 1$. Near either wall, the curvature becomes large and the Newton step in that coordinate is automatically damped.

14 Difference from modern practical IPM

The method proved above is a short-step feasible-start IPM. It is designed to make the convergence proof clean. Practical solvers usually use a different engineering design.

14.1 Short-step theoretical IPM

The proof above uses the following choices.

1. It starts from a strictly primal-dual feasible point.
2. It keeps exact primal and dual equality feasibility at every step.
3. It explicitly maintains the central neighborhood $\mathcal{C}_\gamma$.
4. The centering parameter σ is fixed.
5. The step size is chosen conservatively so that the proof can control all second-order product errors at once.

This gives a transparent worst-case guarantee, but the steps are often much smaller than what one would take in practice.

14.2 Modern practical IPM

Modern implementations care more about total running time and numerical robustness. They often use predictor-corrector directions, adaptive centering, fraction-to-boundary step sizes, infeasible-start formulations, presolve, scaling, sparse factorization, iterative refinement, and crossover. Mehrotra-style predictor-corrector methods became a central practical design for LP; implementations also handle bounds on primal variables directly. Lustig, Marsten, and Shanno describe such an implementation and report performance improvements on NETLIB instances. Gondzio's survey emphasizes that modern IPM implementations combine polynomial-time theory with extensive numerical linear algebra and preprocessing techniques.

Therefore, the proof in this note should be read as a clean theoretical core, not as a full implementation recipe for a commercial LP solver.

14.3 Connection to IPM-MPNN

Qian, Chetelat, and Morris encode LP instances as graphs and show that message-passing neural networks can simulate standard IPM iterations. They also emphasize the use of MPNNs as lightweight proxies adapted to a distribution of LP instances. The short-step method in this note is useful for such a theory because each iteration has a clear algebraic structure: local complementarity products, global averaging through μ , and a Newton system tied to the constraint matrix A .

For the box-constrained version, a graph-based simulation should track at least the variable-node quantities

$$x_i, \quad q_i = 1 - x_i, \quad s_i, \quad k_i,$$

the constraint-node quantities

$$r_j = (Ax - b)_j, \quad w_j,$$

and the global quantity

$$\mu = \frac{x^\top s + q^\top k + r^\top w}{2n + m}.$$

15 Final summary

The convergence proof can be remembered in one sentence:

A short-step IPM keeps all slack-multiplier products balanced, uses a Newton direction to move their average from μ toward $\sigma\mu$, and takes a small enough step so that every product remains positive and balanced.

For the box-constrained LP, the products are

$$x_i s_i, \quad (1 - x_i)k_i, \quad (Ax - b)_j w_j.$$

The average product is

$$\mu = \frac{x^\top s + (1 - x)^\top k + (Ax - b)^\top w}{2n + m}.$$

The normalized variables

$$u_i = \Delta z_i \sqrt{\lambda_i / z_i}, \quad v_i = \Delta \lambda_i \sqrt{z_i / \lambda_i}$$

are introduced because they simultaneously encode

$$u_i + v_i = \frac{\sigma\mu - z_i \lambda_i}{\sqrt{z_i \lambda_i}}, \quad u_i v_i = \Delta z_i \Delta \lambda_i.$$

The identity $U^\top V = 0$ makes the second-order errors globally controllable. This yields a safe step of size $\Omega(1/(2n + m))$, and hence

$$\mu_t \leq \exp\left(-\Theta\left(\frac{t}{2n + m}\right)\right) \mu_0.$$

Finally, because the duality gap is exactly $(2n + m)\mu_t$, the primal objective value converges to the LP optimum.

References

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